

Agency Before Agents- A Mandate to Augment Human Agency in Enabling Relationship Centered Care

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CCS CONCEPTS • **Applied computing** → Life and medical sciences → Health informatics; • **Human-centered computing**
~ Human computer interaction (HCI)

Additional Keywords and Phrases: Human Agency, Generative AI, Relationship-Centered Care, Agentic Automation, Caregiver burnout, Clinician burnout.

ACM Reference Format:

Ajeet Singh. 2026. Agency Before Agents- A Mandate to Augment Human Agency in Enabling Relationship Centered Care. In Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems, April 16th 2026, Barcelona, Spain. ACM. New York, NY, USA. 6 pages. <https://doi.org/XXXXXXX.XXXXXXX>

1 INTRODUCTION

The drive towards automation of labor and human processes has progressed dramatically from the Taylorism of the early 20th century. Computer software has assumed aspects of managerial functions over workers for decades. For the first time, organizations were able to oversee workers without human intervention, building new practices under the system of algorithmic management [22]. The datasets used to train algorithms, especially after the advent of transformer architecture, continue to grow exponentially larger over time, tripling in size each year since 2010 [36]. In the three years since the release of OpenAI’s ChatGPT 3, we have seen a proliferation of generative language and multimedia models which enabled individuals to produce synthetic content and is purported to help them automate specific actions and workflows. In healthcare, such generative AI models have found applications across triage and clinical workflows, medical research and education, billing and administration, and more [23]. In the drive to further augment these capabilities, AI companies pioneered new models with the intended capability to execute a series of complex goals on behalf of users. These tools, known as agents, may pursue lengthy open-ended goals, independently producing a plan and taking action to accomplish them, ostensibly with minimal human oversight and intervention [20].

AI agents are built on principles of optimization, but in contrast to the early tools of algorithmic management, the architecture of contemporary models obfuscates their behavior and methodology [13]. Their underlying LLMs use hundreds of billions of parameters built to predict words in sequence, and demonstrate impressive capabilities in mimicking human language and tasks. With additional layers of fine tuning, filters, system prompts, and more, developers steer LLM ‘behavior’ to more closely align with user values and accomplish such goals [14]. Yet, LLMs may demonstrate their own emergent preferences and behavioral biases, and actively produce outputs to conceal such behavior [25,30,31].

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Manuscript submitted to ACM

In clinical settings, the weight of these decisions can cost patients their lives, and significant progress remains in addressing the risks that patients, caregivers, and providers are exposed to by these models. The framing of relationship-centered care (RCC) can help in catalyzing new practices in strengthening human relations in clinical settings. To underline this framing, I propose orienting methodologies to seriously consider the nature of human agency. Specifically, I call for practices in the design, deployment, and evaluation of algorithmic models to characterize and measure how a model will materially enhance the human agency experienced by patients, caregivers, and providers. By embedding a mandate to augment human agency, I argue that any approach to RCC that strips or fails to preserve their sense of agency cannot meaningfully enhance the human connections in healthcare, and must not advance to the clinical setting.

2 FOUNDATIONS FOR AGENTIC MODELS

AI industry executives have long praised the abilities of their generative AI tools, heralding the omniscient agents that will ‘benefit all of humanity’ [29,38]. However, what do we already know about the technologies upon which they’re built? We can point to substantial research on the performance of learning algorithms to identify serious questions about risks, safety, and equity. A well-intentioned risk prediction algorithm deployed to improve population health management reinforced racial biases, reducing resource allocation to sick black patients compared to white patients, while more benign-appearing attempts to improve clinic scheduling by predicting no-shows overestimated its accuracy, risking vulnerable populations to double-booked appointments and less time for providers to address their needs [24,28,34].

The experience with large language models (LLMs) dramatically amplified concerns, as their outputs have long been demonstrated to reinforce biases along racial, ethnic, and gendered lines, magnifying hegemonic perspectives from training data [2]. Trained to pass tests, LLMs are incapable of conceptualizing truth, or any concept for that matter- the errors, or ‘hallucinations,’ are inevitable results of the statistical process for producing plausible outputs [19].

Even if performance challenges are mitigated, the use of algorithms to intermediate the relationship between an actor and other people, decisions, or information fundamentally changes the relationship between the two. For example, the incorporation of algorithms in newsrooms dramatically changed editorial decision-making and journalist practices in divergent ways across international contexts, while their use also changed how electoral campaigns participated in democratic practices, leveraging misinformation and disinformation in ways that have dismantled once-shared epistemologies [6,12].

The impacts of algorithms on our relationships may not immediately be self-evident. Imbued with the politics of their designers, algorithms reflect decisions on values, beliefs, and ethics [39]. These decisions, whether made implicitly or explicitly, result in certain ideas, objects, and people receiving greater visibility and importance than others [1,17]. From credit scoring applications that determine loan eligibility, to recommender algorithms that maximize engagement on social media platforms, algorithms process their training data to classify, order, and recommend outputs in ways that reflect the politics of the people and systems in which they were built and deployed [3,4,32]. In many situations, this means reproducing histories of biases or prioritizing metrics that introduce new harms to communities.

These impacts are further compounded when algorithms like LLMs are built into agentic systems. When agents produce code to take independent action, any errors produced can compound recursively and invisibly until cascading failures

manifest in additional real world harms [35]. If this is the foundation upon which we are to consider pathways that may support better quality care and health outcomes, how can we advance the field to strengthen human relationships?

3 ELEVATING HUMAN AGENCY

Building RCC practices for the field of AI rightfully elevates questions pertaining to trust, transparency, and relationships. What makes a physician trustworthy? In one philosophical lens, patients may see physicians as trustworthy because of their training, acquired knowledge of medicine, and practiced abilities to deliver care in certain situations. This is considered reliance on the trustee, but to genuinely trust the physician carries the normative expectation that the physician will do the right thing, imposing the moral duty to act with beneficence [10]. For AI systems, demonstrations of transparency can help establish reliability, to enable trust for more effective human-AI interactions [7,26]. These attempts to convey LLM and agentic methodologies and translate them in digestible ways for patients to understand are valuable and necessary areas for investigation. However, as we think about the notion of reliance, it calls for an equally deep investigation of the agent’s output and impact.

A sense of agency is the human’s subjective awareness of initiating, executing, and controlling one’s own volitional actions [15,18]. The introduction of automated systems into human processes shifts the human’s involvement away from manual control to decreasing levels of supervision- deployments of automated systems have been shown to decrease participants’ sense of agency in some settings, while augmenting it in others [8]. A decreased sense of agency often threatens the acceptability of the automated process’ decisions by humans- an “out-of-the-loop” problem where the participant feels helpless, unable to understand the intentions or predict the outcome of actions of the system [21]. Furthermore, a sense of agency is needed to help attribute any causal and moral responsibility to the actions occurring, a loss of which could lead to a moral detachment of the individual from the agent’s actions.

For patients, caretakers, and providers to feel as though a new tool is reliable, trustworthy, and effective, they must be able to preserve some functional sense of agency in which they feel as though they are in control of and responsible for their actions. In contrast to earlier rules-based algorithms, where explicit programming produced a predictable and deterministic output, probabilistic models like LLMs and Agentic systems will always involve uncertainty. But unlike human thought, the decision-making processes often remain inaccessible. While the field of explainable AI emerged to help make the behavior of AI systems interpretable, for the public to gain confidence in the rationality of outputs, this is an insufficient answer to the pervasive and considerable debt of human agency [33].

The impacts of agentic automation require both quantitative and qualitative approaches to understand and rebuild the human experience of agency. Patients have long suffered the greatest insults to their agency when faced with medical needs, surrendering themselves to complex and opaque systems of care delivery [37]. Caregivers struggling with burdens of grief, guilt, isolation, and more also face a considerable loss of agency that carries over into their ability to care for themselves and other obligations [9]. For physicians, the burnout epidemic is rooted deeply in feeling a lack of agency as well [16]. In these settings, meaningful relationships among patients, caregivers, and providers still occur in spite of suffocating circumstances. In studying drivers of authentic connections in provider relationships, there needs to be a mutual participation in care, shared decision-making with an appreciation for the patient’s agency in their own health and healing [11]. This is just as true between patients and their caregivers, where families may not be prepared to practice informed decision-making when playing a crucial role supporting recipients facing debilitating health challenges [27]. Preserving

human agency preserves the space to practice compassion and empathy, for trust and respect to cultivate a shared sense of agency to drive better care delivery and support.

4 A POLITICS OF REFUSAL

To build and govern technologies in healthcare, we cannot interrogate these tools as simply technical elements solving technical challenges. They exist embedded within and in relation with society as a singular coherent system. This sociotechnical understanding recognizes that the performance and impacts of technologies are results of both social dynamics and technical designs. They are inextricable from each other [5].

As such, we must recognize that these models are not deployed in isolation- a critical analysis of the context for the care settings in which interactions occur must also consider the relationships and tensions that exist outside of care settings. These are the worlds that shape the thoughts, stressors, and priorities of our patients who must continue living with their health after leaving the clinic, for caregivers who struggle with the burdens of their own lives as they try to help carry the burdens of others, and for providers who must navigate the hurdles and constraints of healthcare as they strive to practice medicine.

Relationship-Centered Care will remain a distant ambition if attempts at automation are not designed with an intention to augment the sense of agency for patients, caregivers, and providers. As clinicians, technologists, researchers, and compassionate members of society, we must therefore practice a politics of refusal when faced with tools that will further extract agency from these groups. We must set these boundaries firm, and build on histories of design knowledge to innovate new tools to make RCC a reality in the world of AI.

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